

Tooba IMTIAZ

✉ imtiazt@northeastern.edu | 🔗 toobaimt.github.io | [in](#) | [🎓](#)

EDUCATION

FALL 2021 - PRESENT	PhD Candidate , ELECTRICAL ENGINEERING, <i>Northeastern University</i> , Boston	Advisor: Prof. Jennifer Dy
2018 - 2020	Masters , ELECTRICAL ENGINEERING, (GPA: 3.87/4.3) <i>Korea Advanced Institute of Science and Technology (KAIST)</i> , S. Korea	Advisor: Prof. In-So Kweon
2014 - 2018	Bachelors , ELECTRICAL ENGINEERING, (GPA: 3.93/4.0, Rank: 5 th /156) <i>National University of Sciences and Technology (NUST)</i> , Pakistan	Advisor: Prof. Faisal Shafait

WORK EXPERIENCE

MAY 2026 - PRESENT	PhD Intern WAYMO , PERCEPTION TEAM <i>Mountain View, CA</i> Working on sensor fusion under compromised sensing conditions.
JUNE - NOV 2025	Student Researcher GOOGLE , PIXEL BIOMETRICS AI RESEARCH (BAIR) <i>Seattle, WA</i> Developed a text-conditioned multi-view latent diffusion model for stylized 3D scene generation.
SEP 2024 - MAY 2025	Student Researcher GOOGLE BEAM (PROJECT STARLINE) <i>Cambridge, MA</i> - Continued research as an extension of my internship project for Google Beam. - Proposed a feed-forward, generalizable 3DGS-based novel view synthesis framework capable of reconstructing wide-coverage, high-resolution scenes, achieving state-of-the-art performance in a single inference pass. ("LVT", SIGGRAPH Asia 2025)
MAY - AUG 2024	Research Intern GOOGLE BEAM (PROJECT STARLINE) <i>Playa Vista, CA</i> - Worked on efficient 3D reconstruction and novel-view synthesis for Google Beam. - Proposed feed-forward architecture achieved plausible quantitative and qualitative results, despite its simplicity.
SEP 2021 - PRESENT	Research Assistant MACHINE LEARNING LAB @ SPIRAL , <i>Northeastern University, Boston</i> - Developing a novel view synthesis framework to visualize previously unseen depths in 3D reflectance confocal microscopy (RCM) images of human skin, enabling early detection of skin diseases and cancers. - Member of the Northeastern PAIR (Physical AI Research) initiative for developing a physics-aware world model - Contributed to a regularization-based approach for improving continual learning. ("STAR", ICLR 2025) - Developed an optimization-based sparse adversarial attack on images and evaluated its interpretability. ("SAIF", TMLR 2025) - Implemented NeRF-based 3D scene reconstruction from phone camera videos to facilitate at-home patient health monitoring. - Formulated a clustering-based loss to improve the performance of 3D-object detection from multiview 2D inputs.
SPRING 2023	Teaching Assistant EECE7397 Advanced Machine Learning, Northeastern University
SEP 2020 - AUG 2021	Consultant - ML and AI ENDRESS+HAUSER , <i>Maulburg, Germany</i> Proposed ML and CV-based solutions for process automation and optimization. Led two projects, both deployed to production: Deep learning for unsupervised 3D classification: used Autoencoders, Capsule architectures, and Implicit Neural Networks. Forecasting on time series: utilized DNNs and Temporal Transformers to predict compound concentrations in liquids using sensors measuring base physical quantities. Achieved ~ 96% accuracy w.r.t. specialized physical sensors.
SEP 2018 - AUG 2020	Research Assistant ROBOTICS AND COMPUTER VISION LAB , <i>KAIST, South Korea</i> Bosch-RCV Project: Performed camera calibration, data collection, and vehicle trajectory estimation. Designed an occlusion-robust vehicle re-identification method using GANs for seamless tracking across a multi-camera surrounding awareness system. Universal Adversarial Perturbations: Developed novel adversarial attack algorithms. Published at CVPR, AAAI, and ACCV '20.
SEP 2015 - MAY 2018	Research Intern TUKL-NUST R&D CENTRE , <i>NUST, Pakistan</i> - Proposed table detection and parsing in document images using ML and CV (LSTMs, text classification, clustering algorithms). - Implemented LSTMs for handwritten address recognition to sort postal mail.

PUBLICATIONS

CD-RCM: Generalizable Continuous-Depth Novel View Synthesis for Reflectance Confocal Microscopy | **Preprint 2026**

T. Imtiaz, M. Rajadhyaksha, K. Kose, J. Dy

Human Cognition in Machines: A Unified Perspective of World Models | **Preprint 2026**

T. Rupprecht*, P. Zhao*, A. Taherin*, A. Akbari*, A. Akbari*, Y. He*, T. Imtiaz*, et al.

PanoWorld: Geometry-Consistent Panoramic Video World Modeling | Preprint 2026

L. Jiang, X. Bai, B. Galoaa, S. Moezzi, C. J. Lee, T. Imtiaz, E. Yeh, J. Dy, Y. Wang, S. Ostadabbas

LVT: Large-Scale Scene Reconstruction via Local View Transformers | SIGGRAPH Asia 2025

T. Imtiaz*, L. Chai*, K. Heal, X. Luo, J. Park, J. Dy, J. Flynn

STAR: Stability-Inducing Weight Perturbation for Continual Learning | ICLR 2025

M. Eskander, T. Imtiaz, D. Hill, Z. Wang, J. Dy

ADAPT to Robustify Prompt Tuning Vision Transformers | TMLR 2025

M. Eskander, T. Imtiaz, Z. Wang, J. Dy

SAIF: Sparse Adversarial and Imperceptible Attack Framework | TMLR 2025

T. Imtiaz, M. Kohler, J. Miller, Z. Wang, M. Eskandar, M. Sznaier, O. Camps, J. Dy

Volumetric Propagation Network: Stereo-LiDAR Fusion for Long-Range Depth Estimation | IEEE RA-L 2021

J. Choe, K. Joo, T. Imtiaz, I.S. Kweon

Understanding Adversarial Examples from the Mutual Influence of Images and Perturbations | CVPR 2020

C. Zhang*, P. Benz*, T. Imtiaz, I.S. Kweon

CD-UAP: Class Discriminative Universal Adversarial Perturbation | AAAI 2020

C. Zhang*, P. Benz*, T. Imtiaz, I.S. Kweon

Double Targeted Universal Adversarial Perturbations | ACCV 2020

P. Benz*, C. Zhang*, T. Imtiaz, I.S. Kweon

Data from Model: Extracting Data from Non-robust and Robust Models | CVPRW 2020

P. Benz*, C. Zhang*, T. Imtiaz, I.S. Kweon

Universal Adversarial Perturbations are Not Bugs, They are Features | CVPRW 2020

P. Benz*, C. Zhang*, T. Imtiaz, I.S. Kweon

PATENTS

3D Novel View Synthesis from Sparse Optical Sectioning Images | Pending

T. Imtiaz, K. Kose, J. Dy, M. Rajadhyaksha

A feedforward 3D synthesis method for generating continuous tissue views from sparse optical sections.

LVT: Large-Scale Scene Reconstruction via Local View Transformers | Pending

J. Flynn, L. Chai, T. Imtiaz

A generalizable novel view synthesis framework for reconstructing wide-coverage, high-resolution scenes, achieving state-of-the-art performance.

SCHOLARSHIPS AND AWARDS

2020	Qualcomm Innovation Fellowship Award, South Korea (among the 20 awardees)
2014-2018	NUST Merit Scholarship (Awarded to top-3 GPA holders of cohort)
2017	Global UGRAD Exchange Program, US Dept of State (~ 7.6% selection rate)

SKILLS

PYTHON	JAX, PyTorch, Tensorflow, Keras, Numpy, scikit-learn, cuda, Matplotlib
C / C++ / JAVA	Object-oriented programming, Data structures, frontend and backend dev
MISC.	MATLAB, Unix, gcc, Git, SQL, TeX LaTeX, ROS, AutoCAD

ACADEMIC SERVICE

WORKFLOW CHAIR	AAAI 2024 - Managed the AAAI 2024 paper review process for 12,100 submissions, working with 7k reviewers, 765 senior program committee (SPC), and 320 area chairs (AC). - Used topic modeling and text similarity to determine reviewer, SPC, and AC assignments.
CONFERENCE REVIEWER	NeurIPS 2025-26, ECCV 2024 & 26, ICLR 2025-26, AAAI 2026, CVPR 2024, ACCV 2024, ICCV 2023.
VOLUNTEER	NeurIPS 2025, ICML 2022.